

Project Topics Specification

1. Processing in Memory (PIM) or Near Memory Computing (NMC)

A project should include ISA design related to PIM/NMC, compiler, simulation of hardware (memory & computing units) and execution, benchmarking, and sample applications.

Bonus points will be given if digital logic circuit simulation is included.

2. Integrated Compute and Storage (ICC) Architecture

Simulation of hundreds or more interconnected nodes of ICC, that can make vector/tensor or machine learning computation, and performance comparison with traditional GPU/CPU computations.

3. Companion Robot built on Raspberry Pi device

By installing an AI LLM model on a Raspberry Pi device, to offer audio/text conversation and other companion robot capabilities.

Bonus points will be given if video is also supported.

4. High Bandwidth Memory (HBM)

Using a memory simulator to simulate different memory architecture, such as HBM, 2.5D memory, etc., and compare the performance of different memory types for sample applications.

5. Simulation of Optical Computing to Support Vector/Tensor Computation

Select or design a simulation environment of optical computing to support vector/tensor computation, or even matrix computation and AI applications.

6. Evaluation of On Device LLMs

Evaluation and benchmarking of on device large language models such as Phi4, MiniCPM, SmolLM, Qwen, Llama etc.

7. Your own project related to computer architecture

The project should involve hardware, simulation, application, and performance comparison...